FundFlow — Accounting & Reconciliation Reference

Accountant one-pager · Print or use as a 7-slide deck

Slide 1 — Chart of accounts (tenant)

Account	Type	Normal balance	Pool mirror
Master Bank	Asset	Debit	—
Master Cash	Asset	Debit	Σ member cash
Master Fund	Equity/pool	Credit	Σ member fund
Member Cash	Asset (per member)	Debit	master cash
Member Fund	Equity (per member)	Credit	master fund
Loan sub-account	Contra / tracking	Varies	Per loan
Master Fees	Revenue/reserve	Credit	Master only

Slide 2 — Pool invariants (nightly)

Code	Check
`MASTER_CASH_POOL_DRIFT`	`master_cash.balance` Σ `member_cash.balance`
`MASTER_FUND_POOL_DRIFT`	`master_fund.balance` Σ `member_fund.balance`
`MEMBER_CASH_DRIFT`	Component formula vs stored cash balance
`MEMBER_FUND_DRIFT`	Component formula vs stored fund balance

Mirroring is suppressed mid-transaction (`masterPoolMirrorInProgress`) — realtime checks skip during paired postings.

Slide 3 — Standard journal patterns

Contribution collection

DR	CR
Member cash	
Master cash (mirror)	
	Member fund
	Master fund (mirror)

Loan disbursement

DR	CR
Member fund (portions)	
Master fund (mirror)	
Loan account (principal)	
	Member cash
	Master cash (mirror)

Cash-out approval

DR	CR
Member cash	
Master cash (mirror)	

Plus uncleared `bank_transactions` row — clearance updates flags only.

Loan repayment (import / EMI)

Step	DR	CR
Cash-in mirror	Member + master cash	
Collection		(paired debit same accounts)
Fund leg		Member + master fund
Principal		Loan sub-account

Slide 4 — Bank clearance vs ledger

Concept	Ledger	Bank clearance
Deposit accept	CR cash (intent)	Uncleared line → match import
Cash-out accept	DR cash (intent)	Uncleared line → match import
Mirror to cash	Master bank master cash	Statement status change
Clear / Match	**No additional legs**	`is_cleared` linkage

Do not pass `member_id` on master cash legs unless documented — triggers `onMemberCashIncreased()` auto-collection.

Slide 5 — Loan funding split

At disbursement, `member_portion` + `master_portion` = `amount_approved`.

Portion	Economic meaning	Repayment obligation
Member portion	Equity already in pool	Not collected via EMI
Master portion	Pool exposure	Collected via EMI
Settlement threshold	% of approved (e.g. 16%)	Collected via EMI

Fully member-funded (`master_portion` = 0, `settlement_threshold` = 0): no EMI schedule; loan completed at import. Principal applied at disbursement via fund debit + loan account credit.

Repayment target: `master_portion + (amount_approved × settlement_threshold)`.

Slide 6 — Legacy migration notes

Issue	Cause	Remediation
Cash balance txn net	Repair deleted txs without balance adjust	`php artisan accounting:rebuild-balances --tenants=…`
Phantom cash after reclassify	`reverseContributionPrincipal` + raw `delete()`	Fixed in `ContributionService` migration repair
Pending EMI on fund-only loan	Import created schedule despite zero obligation	`legacy:rebuild-loan-installments`

Legacy repayments use paired cash mirrors (credit then debit) — net cash zero per payment; fund leg carries economic effect.

Slide 7 — Reconciliation workflow

```
Import bank CSV
    → Mirror to master cash (master bank leg)
    → Post to member (member cash leg)
    → Auto-collection (contributions / EMIs)
    → Nightly invariants
    → Investigate exceptions → manual correction journals
```

Key services: `AccountingService` (mirror helpers), `FundPostingService`, `LoanLedgerService`, `MemberCashOutService`, `MemberInvariantService`, `ReconciliationCorrectionService`.

References: `docs/fund-flow-dynamics.md` · `docs/fund-flow-reconciliation-and-scheduler.md` · `.cursor/rules/accounting-master-member-sync.mdc` · `docs/collection_cycle_workflow.md`

For component-level cash formula see `MemberInvariantService` §5.13.